

Zhytomyr Agrotechnical Professional College

ICE Design 0.8.6 · Teaching guide

Vehicle powertrain design calculation

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1. Introduction and problem statement

How to read the notation

Symbols follow full quantity names: power (P), torque (M), rotational speed (n), vehicle speed (v), mass (m), force (F), energy (E), voltage (U), current (I), resistance (R), and efficiency (η). Lowercase p denotes pressure. A subscript identifies the component or operating mode, for example, P_{S1} — continuous power, M_{peak} — peak torque.

An internal combustion engine (ICE) may use spark ignition (SI) or compression ignition (CI). PFI means port fuel injection; DI means direct injection, and GDI means gasoline direct injection. The slider-crank mechanism and valve gear are distinguished; TDC and BDC mean top and bottom dead centre, and CA means crank angle.

BEV means battery electric vehicle; HEV means hybrid electric vehicle; PHEV means plug-in hybrid electric vehicle. P2 is a parallel layout with an electric machine between the engine and gearbox. The battery's SOC is its state of charge; OCV is open-circuit voltage (U_{oc}). Equivalent battery internal resistance is R_{int}; this is an ohmic internal-resistance model. NMC means lithium-ion nickel manganese cobalt oxide; LFP means lithium iron phosphate, $LiFePO_4$. Chemistry alone does not determine pack voltage, specific energy, or current limits.

PMSM means permanent-magnet synchronous motor. d and q are the direct and quadrature axes of a rotating coordinate system; L_{d} and L_{q} — axis inductances; ψ is flux linkage. RMS means root mean square; for a sinusoid, peak amplitude equals √2 times RMS. In phase voltage, peak denotes amplitude; in power/torque characteristics, it denotes short-duration duty. DC means direct current; AC means alternating current. S1 is continuous duty under steady thermal conditions; peak is short-duration peak duty. SVPWM means space-vector pulse-width modulation.

WLTP means Worldwide harmonized Light vehicles Test Procedure; its driving cycle is WLTC, Worldwide harmonized Light vehicles Test Cycle. PER means pure electric range. UBE means usable battery energy, measured here at the terminals; energy consumption per distance is ($\mathrm{EC}_{\mathrm{DC}}$). CSS means constant-speed segment; M and E denote intermediate and final segments. The index in WLTC_1, WLTC_2 denotes the repetition number, not another cycle class.

EPA is the US Environmental Protection Agency. ALPHA means Advanced Light-duty Powertrain and Hybrid Analysis. UDDS is the Urban Dynamometer Driving Schedule; HWFET is the Highway Fuel Economy Test; US06 is the aggressive supplementary schedule. These names identify specific speed profiles.

ECU means electronic control unit; BMS means battery management system; OBD2 means second-generation on-board diagnostics. RMSE means root mean square error. The International System of Units (SI) is distinguished by context from SI meaning spark ignition. File formats are PDF (Portable Document Format), CSV (Comma-Separated Values), JSON (JavaScript Object Notation), SVG (Scalable Vector Graphics), HTML (HyperText Markup Language); ZIP is an archive format.

This is a general design and verification calculation for an engine and powertrain. First specify the required performance and adopt the parameters, then check the processes, dimensions, and subsequent calculations.

Selecting a petrol engine, diesel engine, hybrid, or BEV changes the active fields and chapters. The carburettor branch is excluded. Modern injection and boosting are represented by the available model; Euro 5–6 emissions compliance is not numerically verified.

P_{req} is a requirement; P_{calc} is a result. Calculated and adopted geometry are stored separately. Changing the inputs makes all dependent results stale until recalculation.

2. Design inputs

Course project assignment for Automotive Engines. The principal inputs follow the assignment form; hybrid and electric powertrains additionally specify the system architecture and component requirements.

Supporting assumptions appear beside their respective calculations in subsequent chapters. The initial dataset is filled in; all applicable parameters can be edited.

3. Calculation assumptions and initial dimensions

The elemental C/H/O composition and oxygen mass fraction in air of 0.23 determine theoretical air demand. The effective heat capacities c_{v}(T)=675+0.25T and R=287 are not chemical-equilibrium properties of the products.

In design mode, the initial p_{e0} determines only the first approximation to D. The iterative solver then finds D at the specified S/D from the brake power of the reintegrated cycle. In verification mode, D and S are specified.

The calculated bore and stroke are not nominal drawing dimensions. For preliminary adoption, D and S may be rounded upwards using increment I-177, normally 1 mm, and the calculation repeated. Production engines need not use integer millimetres: BMW specifies D = 82 mm and S = 94.6 mm [bmw-bore-stroke].

Supporting engine-speed profiles

These values are adopted design guidelines, not a universal classification or speed limiter. Refine recommendations using the selected engine's specifications. In a series hybrid, engine speed is independent of traction-machine speed.

Table 3.1. Initial speeds and intervals by engine profile.

Profile	Initial n, min⁻¹	Characteristic interval, min⁻¹
M1 (passenger car) · naturally aspirated petrol [2]	5500	800–6500
M1 (passenger car) · boosted petrol	5500	1500–6500
M1 (passenger car) · petrol hybrid profile	5200	1200–6000
M1 (passenger car) · direct-injection diesel [2]	3800	800–4500
N3 (goods vehicle over 12 t) · direct-injection diesel [2]	2100	600–2500
M1 (passenger car) · petrol educational profile, 7000 min⁻¹ [1]	7000	2100–7000
M2 (bus up to 5 t) · naturally aspirated petrol [2]	5000	780–5800
M2 (bus up to 5 t) · direct-injection diesel [2]	3500	780–4000
M3 (bus over 5 t) · naturally aspirated petrol [2]	4000	650–4600
M3 (bus over 5 t) · direct-injection diesel [2]	2200	650–2500
N1 (goods vehicle up to 3.5 t) · naturally aspirated petrol [2]	5000	780–5800
N1 (goods vehicle up to 3.5 t) · direct-injection diesel [2]	3500	780–4000
N2 (goods vehicle 3.5–12 t) · naturally aspirated petrol [2]	4200	700–4800
N2 (goods vehicle 3.5–12 t) · direct-injection diesel [2]	2400	700–2800
N3 (goods vehicle over 12 t) · naturally aspirated petrol [2]	4000	600–4600

No production brand is required. After selecting a category, recommended initial values can be applied; manually entered values are preserved. Required power and the calculated result remain separate quantities.

Occupant and payload guidelines

The category describes purpose and mass limits; you specify trip payload. Initial project values and approximate nominal payload from the Automobile adaptation are given below.

Category	Initial cargo, kg	Reference payload, kg
M1 · Passenger car	200	≈ 546
M2 · Bus to 5 t	500	≈ 800
M3 · Bus above 5 t	1 500	≈ 5 000
N1 · Goods vehicle to 3,5 t	500	≈ 500
N2 · Goods vehicle above 3,5–12 t	2 000	≈ 4 500
N3 · Goods vehicle above 12 t	5 000	≈ 14 000

M1 has up to 8 passenger seats excluding the driver; M2/M3 have more than 8. Tabulated payloads are editable design guidelines, not regulatory limits for every category. Use the actual vehicle's specified payload. [19] [9]

Count occupants and luggage once: if the driver is already included in the adopted kerb mass, do not add the driver to payload again. Changing category must not replace manual I-174.

Angular speed of the specified output shaft.

where n_{req} — Speed at which power I-6 is required, s⁻¹

$$\omega_{\text{req}} = 2 \cdot \pi \cdot n_{\text{req}} \tag{3.1}$$

The result passes to dependent calculations without being entered again.

The specified power and rotational speed determine torque at the rated point. This does not search the full characteristic for maximum torque.

where P_{req} — Required output-shaft power, W; ω_{req} — Angular speed of the specified output shaft, rad/s

$$M_{\text{req}} = \frac{P_{\text{req}}}{\omega_{\text{req}}} \tag{3.2}$$

The result passes to dependent calculations without being entered again.

The elemental mass fractions determine the theoretical air requirement. The oxygen fraction in air, 0.23, is an assumption of this method. [3]

where C — Carbon mass fraction in the model fuel; H — Hydrogen mass fraction; O — Oxygen mass fraction

$$l_0 = \frac{\left(\left(\frac{8}{3} \right) \cdot C + 8 \cdot H - O \right)}{0{,}23} \tag{3.3}$$

The result passes to dependent calculations without being entered again.

Piston area.

where D — Adopted bore, m

$$A_{\text{p}} = \frac{\pi \cdot D^2}{4} \tag{3.4}$$

The result passes to dependent calculations without being entered again.

Swept volume of one cylinder.

where A_{p} — Piston area, m²; S — Adopted stroke, m

$$V_{\text{h}} = A_{\text{p}} \cdot S \tag{3.5}$$

The result passes to dependent calculations without being entered again.

Total engine displacement.

where V_{h} — Swept volume of one cylinder, m³; z — Number of cylinders, 1

$$V_{\Sigma} = V_{\text{h}} \cdot z \tag{3.6}$$

The result passes to dependent calculations without being entered again.

The adopted compression ratio and swept volume determine the geometric clearance volume.

where V_{h} — Swept volume of one cylinder, m³; ε — Geometric compression ratio

$$V_{\mathrm{c}} = \frac{V_{\mathrm{h}}}{(\,\varepsilon - 1\,)} \tag{3.7}$$

The result passes to dependent calculations without being entered again.

Connecting rod centre distance.

where S — Adopted stroke, m; λ — Crank-radius-to-rod-length ratio

$$l = \frac{S}{(\,2 \cdot \lambda\,)} \tag{3.8}$$

The result passes to dependent calculations without being entered again.

4. Intake, compression, combustion, expansion, and exhaust

A single-zone 720° cycle is integrated until a periodic state is reached. Valve flows use a compressible-nozzle model; the geometry of the centred slider-crank mechanism is exact. Two inlet and two exhaust valves use a sin² lift law without overlap.

PFI: the fresh mixture already contains fuel. GDI/diesel: air enters through the inlet, while fuel mass and its injection energy are added separately. Diesel uses an assumed ignition delay and two Wiebe phases; combustion cannot precede fuel injection.

There is no spray, NOx/soot chemistry, knock, or multizone stratification model. Boosted intake/exhaust pressures are prescribed boundary conditions, not turbocharger selection results.

The single-zone (0D) model treats the gas in each cylinder as one homogeneous control volume, with one pressure and one mean temperature at each instant. A single zone does not mean a single cylinder. This simplification allows rapid integration of mass and energy balances and calculation of p–V, work, and power; it does not resolve the flame front, spray, local temperatures, or separate fresh and burnt gas zones. The homogeneous-control-volume principle is described in [cantera-reactors]. This application uses its own simplified heat-capacity and heat-release laws. On indicator diagrams, the dashed line represents atmospheric pressure p_0 = 0.101325 MPa.

At each crank-angle step, the change in working-fluid mass accounts for gas exchange and direct injection.

> m is cylinder mass, kg; θ is angle in degrees; n is rotational speed in min⁻¹; $\dot{m}$ is mass flow in kg/s. The factor 1/(6n) converts seconds to one degree. For PFI, fuel mass is already included in the fresh mixture, so the separate term is zero.
>
> $$\frac{dm}{d\theta} = \frac{\dot{m}_{\mathrm{in}} + \dot{m}_{\mathrm{ex}}}{6n} + \frac{dm_{\mathrm{f}}}{d\theta} \tag{4.1}$$

The first law of thermodynamics is integrated. Heat transfer and mechanical work reduce internal energy; enthalpy flows retain the sign of gas exchange.

> U is internal energy, J; Qc is combustion heat, J; Qw is heat transferred to the walls, J; p is absolute pressure, Pa; V is volume, m³; $\dot{H}$ is enthalpy flow, W; hf,eff is the effective enthalpy of injected fuel, J/kg. The laws and boundary conditions are specified in the fields of this chapter.
>
> $$\frac{dU}{d\theta} = \frac{dQ_{\mathrm{c}}}{d\theta} - \frac{dQ_{\mathrm{w}}}{d\theta} - p \cdot \frac{dV}{d\theta} + \frac{\mathrm{H}_{\mathrm{in}} - \mathrm{H}_{\mathrm{out}}}{6n} + h_{\mathrm{f,eff}} \cdot \frac{dm_{\mathrm{f}}}{d\theta} \tag{4.2}$$

The effective-gas equation determines pressure for every calculated state. The maximum below is the maximum in the complete cycle array, not a separately substituted specification value.

R = 287 J/(kg·K) is the assumed effective gas constant; T is the single-zone mean temperature, K. This is not a chemical-equilibrium model of combustion products.

$$p\left(\theta\right) = \frac{m\left(\theta\right) \cdot R \cdot T\left(\theta\right)}{V\left(\theta\right)} \tag{4.3}$$

The complete indicator-diagram loop gives the indicated work of one cylinder; the low-pressure gas-exchange loop retains its own sign.

W_{i} is work per four-stroke cycle, J. Numerical tests compare p–V quadrature against an independent torque integral.

$$W_{\mathrm{i}} = \oint p dV = \int_{0}^{720} p\left(\theta\right) \cdot \frac{dV}{d\theta} d\theta \tag{4.4}$$

For the two-stroke case, exhaust opens first after the power stroke, followed by scavenging; ports close in the reverse order. [4]

$$A_j(\theta)=A_{j,\max}\cdot\sin^2[\pi(\theta-\theta_{j,\mathrm{open}})/(\theta_{j,\mathrm{close}}-\theta_{j,\mathrm{open}})] \tag{4.5}$$

Area outside the open interval is zero. Integrate work over 0–360°; total cylinder cycles per second are z·n when n is in rev/s, and mean indicated torque is $W_i \cdot z/(2\pi)$. Scavenging pressure, angles, and effective areas are assumptions; the blower drive requires separate calculation.

At every step, the integrator applies mass conservation and the first law of thermodynamics. For four-stroke, θ=0° is TDC before intake and the cycle spans 720°; for two-stroke, θ=0° is power-stroke TDC and the cycle spans 360°. Two-stroke exhaust/scavenging ports and reservoir pressure are specified separately. Blower work is excluded. Single-zone temperatures are mean values. Prescribed injection and heat-release laws are not automatic ECU calibration.

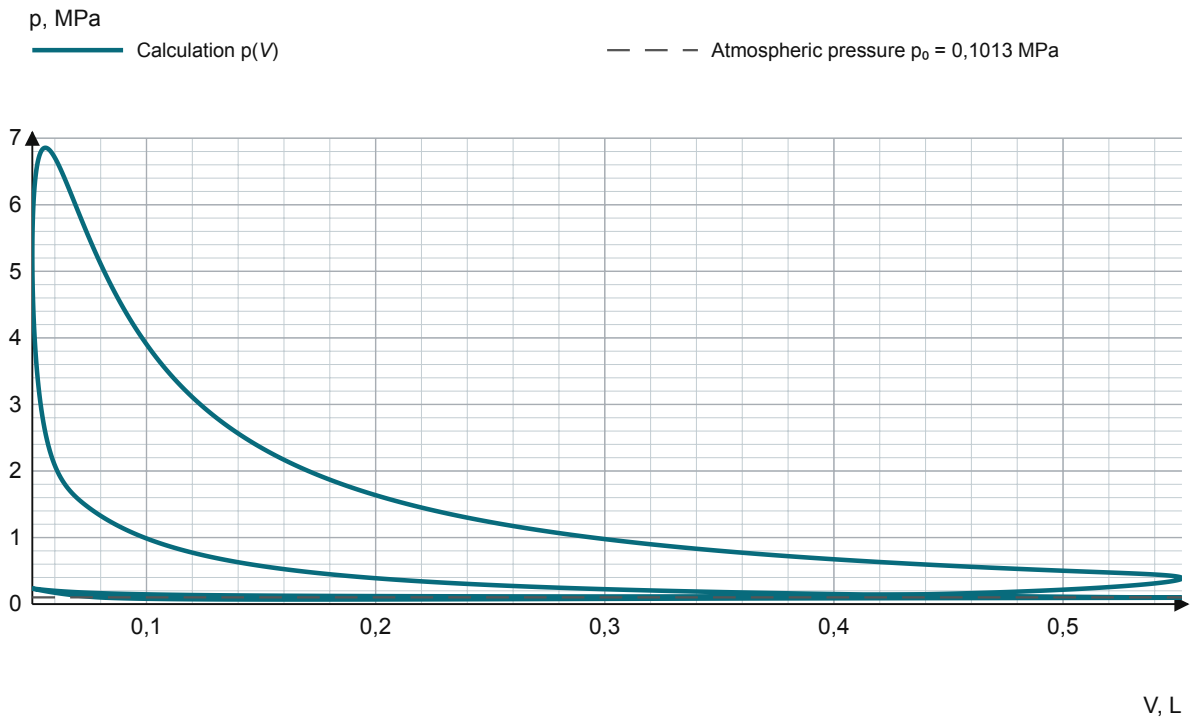


Illustration 4.1 — Indicator diagram p–V. Petrol PFI · 110 kW / 7000 min⁻¹.

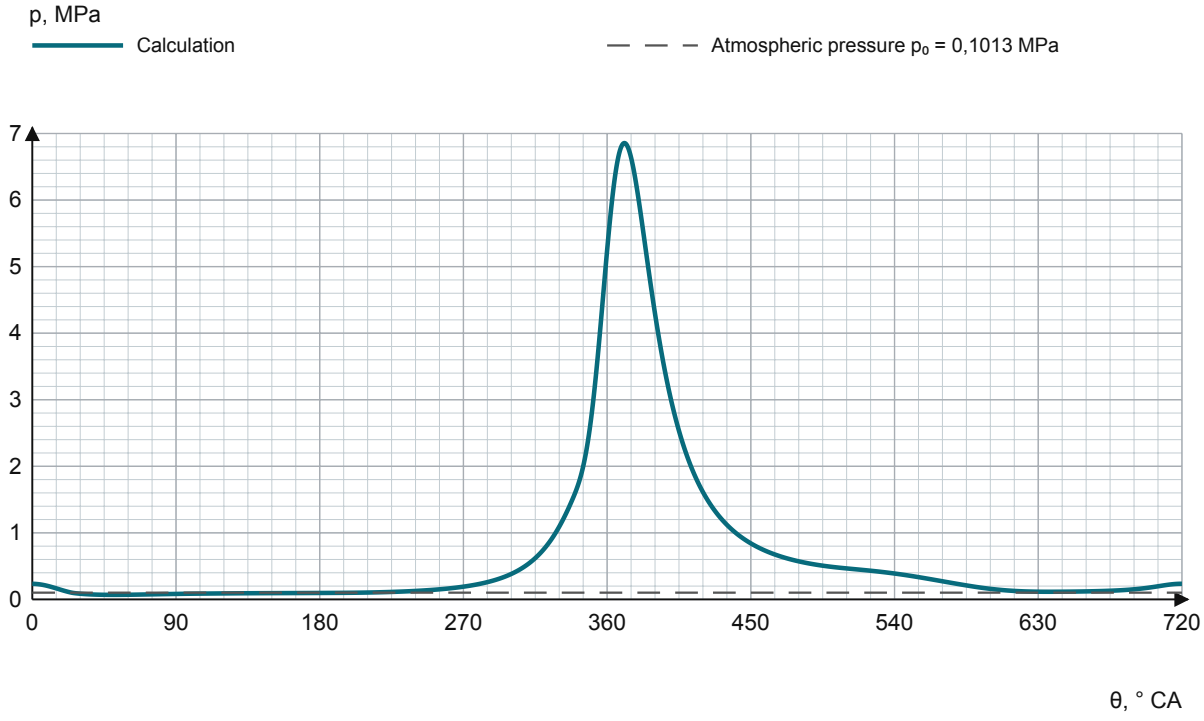


Illustration 4.2 — Crank-angle indicator diagram p–θ, 0–720° CA. Petrol PFI · 110 kW / 7000 min⁻¹.

5. Performance, adopted dimensions, and characteristics

p–V work determines indicated power; mechanical losses are deducted separately. Fuel consumption follows from the cylinder charge and fuel delivery, rather than a brand-specific coefficient.

The speed series is reintegrated at every rotational speed using the same adopted control laws. It is not an optimised manufacturer's full-load characteristic. Load points are explicitly modelled; a missing EPA-map value is not replaced by zero consumption.

The separate Powertrain module accepts numerical packages with component identity, operating envelopes, and data domains. US Environmental Protection Agency (EPA) packages are installed in Data/EPA; each package retains its own provenance and limitations. The characteristic calculated by this Design module remains modelled, not measured.

Crankshaft angular speed.

where n_{req} — Speed at which power I-6 is required, s⁻¹

$$\omega_{\mathrm{e}} = 2 \cdot \pi \cdot n_{\mathrm{req}} \tag{5.1}$$

The result passes to dependent calculations without being entered again.

Each cylinder completes a cycle in two revolutions for a four-stroke engine and one revolution for a two-stroke engine. The total cycle frequency converts J/cycle to W.

where z — Number of cylinders, 1; n_{req} — Speed at which power I-6 is required, s⁻¹; k — Result of the corresponding preceding module

$$\nu_{\mathrm{c}} = \frac{z \cdot n_{\mathrm{req}}}{k} \tag{5.2}$$

The result passes to dependent calculations without being entered again.

Mean piston speed.

where S — Adopted stroke, m; n_{req} — Speed at which power I-6 is required, s⁻¹

$$c_{\mathrm{m}} = 2 \cdot S \cdot n_{\mathrm{req}} \tag{5.3}$$

The result passes to dependent calculations without being entered again.

Indicated mean effective pressure.

where W_i — Indicated work of one cylinder per cycle, J; V_h — Swept volume of one cylinder, m³

$$p_\mathrm{i} = \frac{W_\mathrm{i}}{V_\mathrm{h}} \tag{5.4}$$

The result passes to dependent calculations without being entered again.

Engine indicated power.

where W_i — Indicated work of one cylinder per cycle, J; ν_c — Working cycles of all cylinders per second, s⁻¹

$$P_\mathrm{i} = W_\mathrm{i} \cdot \nu_\mathrm{c} \tag{5.5}$$

The result passes to dependent calculations without being entered again.

Engine mechanical losses.

where p_f — Friction mean effective pressure, Pa; V_h — Swept volume of one cylinder, m³; ν_c — Working cycles of all cylinders per second, s⁻¹

$$P_\mathrm{f} = p_\mathrm{f} \cdot V_\mathrm{h} \cdot \nu_\mathrm{c} \tag{5.6}$$

The result passes to dependent calculations without being entered again.

Modelled mechanical losses are deducted from indicated power. Only then is the result compared with the specified brake power.

where P_i — Engine indicated power, W; P_f — Engine mechanical losses, W

$$P_\mathrm{ICE,calc} = P_\mathrm{i} - P_\mathrm{f} \tag{5.7}$$

The result passes to dependent calculations without being entered again.

Brake mean effective pressure.

where p_i — Indicated mean effective pressure, Pa; p_f — Friction mean effective pressure, Pa

$$p_\mathrm{e} = p_\mathrm{i} - p_\mathrm{f} \tag{5.8}$$

The result passes to dependent calculations without being entered again.

Brake torque at the calculation point.

where $P_{\mathrm{ICE,calc}}$ — Brake power after adopting the dimensions, W; ω_{e} — Crankshaft angular speed, rad/s

$$M_{\mathrm{e}} = \frac{P_{\mathrm{ICE,calc}}}{\omega_{\mathrm{e}}} \tag{5.9}$$

The result passes to dependent calculations without being entered again.

Fuel mass flow.

where m_{fc} — Fuel mass per cycle, kg; ν_{c} — Working cycles of all cylinders per second, s⁻¹

$$m_{\mathrm{f}} = m_{\mathrm{fc}} \cdot \nu_{\mathrm{c}} \tag{5.10}$$

The result passes to dependent calculations without being entered again.

Fuel chemical-energy power.

where m_{f} — Fuel mass flow, kg/s; H_{u} — Lower heating value, J/kg

$$P_{\mathrm{H}} = m_{\mathrm{f}} \cdot H_{\mathrm{u}} \tag{5.11}$$

The result passes to dependent calculations without being entered again.

Brake thermal efficiency — model estimate.

where $P_{\mathrm{ICE,calc}}$ — Brake power after adopting the dimensions, W; P_{H} — Fuel chemical-energy power, W

$$\eta_{\mathrm{e}} = \frac{P_{\mathrm{ICE,calc}}}{P_{\mathrm{H}}} \tag{5.12}$$

The result passes to dependent calculations without being entered again.

Brake specific fuel consumption in SI units.

where m_{f} — Fuel mass flow, kg/s; $P_{\mathrm{ICE,calc}}$ — Brake power after adopting the dimensions, W

$$b_{\mathrm{e}} = \frac{m_{\mathrm{f}}}{P_{\mathrm{ICE,calc}}} \tag{5.13}$$

The result passes to dependent calculations without being entered again.

Volumetric fuel flow.

where m_{f} — Fuel mass flow, kg/s; ρ_{f} — Fuel density, kg/m³

$$V_{\mathrm{f}} = \frac{m_{\mathrm{f}}}{\rho_{\mathrm{f}}} \tag{5.14}$$

The result passes to dependent calculations without being entered again.

The independently calculated power is compared with the requirement. The residual is not forced to zero by assigning the requirement to the result.

where $P_{\mathrm{ICE,calc}}$ — Brake power after adopting the dimensions, W; P_{req} — Required output-shaft power, W

$$\delta \mathrm{P}_{\mathrm{ICE}} = \frac{\left(P_{\mathrm{ICE,calc}} - P_{\mathrm{req}} \right)}{P_{\mathrm{req}}} \tag{5.15}$$

The result passes to dependent calculations without being entered again.

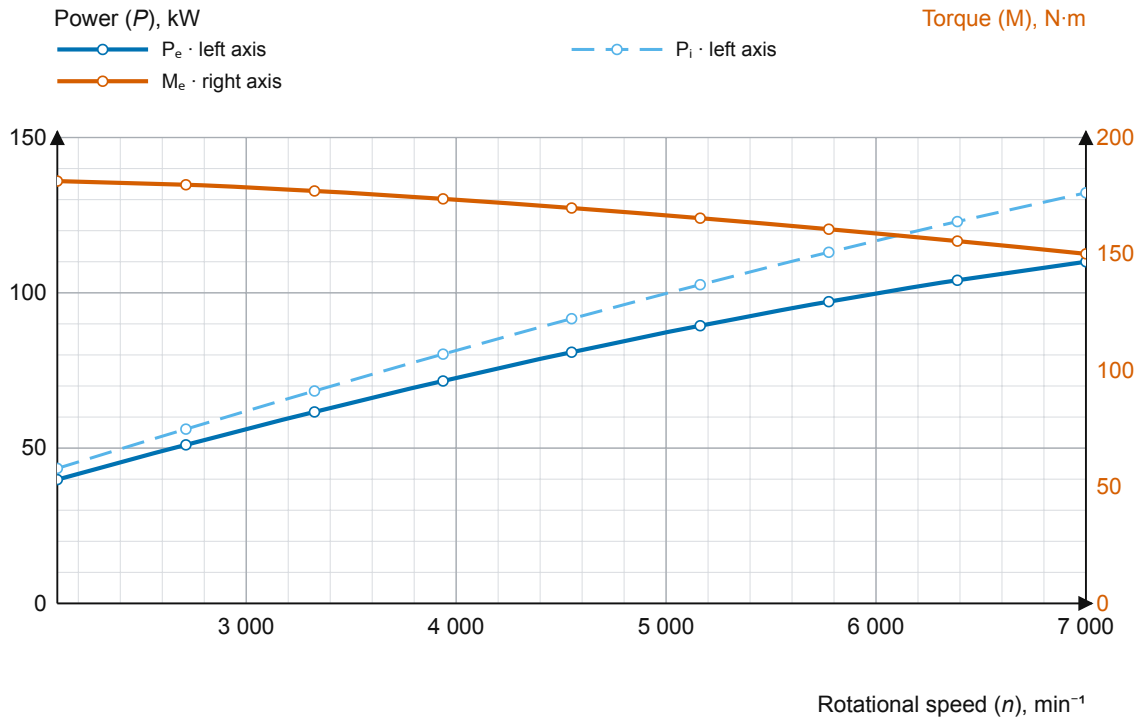


Illustration 5.1 — Full-load speed characteristic: power and torque. Petrol PFI · 110 kW / 7000 min⁻¹.

6. Kinematics, dynamics, and auxiliary systems

Pressure and geometry from the preceding chapters determine gas, inertia, resultant, connecting-rod, and crank forces without manual copying. Total torque assumes evenly spaced firing events over one active cycle: 360°/z or 720°/z.

Energy fluctuation determines the required total inertia, not the final flywheel design. Heat flow gives a preliminary coolant mass flow, not pump head or a production pump. Strength, fatigue, spatial balancing, and bearings are not fully assessed here.

The additional 0.7 assessment uses current forces for preliminary axial/Euler/Goodman, piston-pin, and torsion checks, and heat for a series hydraulic circuit. A separate explicit specification is required; this is not a complete component or fatigue-life calculation. Unknown data are marked not assessed.

The crank radius determines the tangential-force lever arm.

where S — Adopted stroke, m

$$R = \frac{S}{2} \tag{6.1}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

Crankcase pressure is subtracted from cylinder pressure; absolute pressure alone does not give the gas force.

where p — Pressure at the check point, Pa; p_{cr} — Absolute crankcase pressure, Pa; A_{p} — Piston area, m²

$$F_{\mathrm{g}} = \left(p - p_{\mathrm{cr}} \right) \cdot A_{\mathrm{p}} \tag{6.2}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

The inertia force acts opposite to piston acceleration.

where m_{rec} — Equivalent reciprocating mass per cylinder, kg; a — Acceleration at the check point, m/s²

$$F_{\mathrm{j}} = -m_{\mathrm{rec}} \cdot a \tag{6.3}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

The forces are summed algebraically using one sign convention, rather than adding their absolute magnitudes.

where F_{g} — Gas force at the check point, N; F_{j} — Reciprocating-mass inertia force, N

$$F_{\Sigma} = F_{\mathrm{g}} + F_{\mathrm{j}} \tag{6.4}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

The transverse component follows from the current connecting-rod angle.

where F_{Σ} — Resultant axial force, N; β — Connecting-rod angle, rad

$$N_{\mathrm{side}} = F_{\Sigma} \cdot \tan\left(\beta\right) \tag{6.5}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

Force along the connecting rod: FΣ/cos β. An inconsistent expression in the old RTF was not transferred.

where F_{Σ} — Resultant axial force, N; β — Connecting-rod angle, rad

$$S_{\mathrm{rod}} = \frac{F_{\Sigma}}{\cos\left(\beta\right)} \tag{6.6}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

The tangential component produces torque; θ and β are passed as angles.

where F_{Σ} — Resultant axial force, N; θ — Crank angle, rad; β — Connecting-rod angle, rad

$$T = \frac{F_{\Sigma} \cdot \sin\left(\theta + \beta\right)}{\cos\left(\beta\right)} \tag{6.7}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

The radial component loads the bearings; multiplying it by R does not give useful torque.

where F_{Σ} — Resultant axial force, N; θ — Crank angle, rad; β — Connecting-rod angle, rad

$$Z = \frac{F_{\Sigma} \cdot \cos\left(\theta + \beta\right)}{\cos\left(\beta\right)} \tag{6.8}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

Torque is calculated from the tangential force and crank radius.

where T — Crank tangential force, N; R — Crank radius, m

$$M_{\mathrm{c}} = T \cdot R \tag{6.9}$$

The check point and full diagram come from the active cycle (360° or 720°) without entering pressure again.

The integral of total torque over one active cycle equals the work of all cylinders. It is divided by 2πk rad, where k is the revolutions per cycle; mechanical losses have not yet been deducted.

where W_{i} — Indicated work of one cylinder per cycle, J; z — Number of cylinders, 1; k — Result of the corresponding preceding module

$$M_{\mathrm{i,avg}} = \frac{W_{\mathrm{i}} \cdot z}{\left(2 \cdot \pi \cdot k\right)} \tag{6.10}$$

Do not confuse this with mean brake torque $M_e = P_e/\omega$, which already accounts for losses.

The excess-energy fluctuation determines the required total shaft inertia. The inertia of other rotating components must be deducted separately before designing the flywheel.

where $\Delta \mathrm{E}$ — Excess-energy fluctuation, J; δ_{ω} — Allowable speed fluctuation; ω_{e} — Crankshaft angular speed, rad/s

$$J_{\mathrm{req}} = \frac{\Delta \mathrm{E}}{\left(\delta_{\omega} \cdot {\omega_{\mathrm{e}}}^{2}\right)} \tag{6.11}$$

The result passes to dependent calculations without being entered again.

Preliminary required coolant mass flow.

where Q_{w} — Heat flow to the walls, W; P_{f} — Engine mechanical losses, W; c_{c} — Coolant heat capacity, J/(kg·K); ΔT_{c} — Adopted coolant temperature rise, K

$$m_{c} = \frac{\left(Q_{w} + 0{,}7 \cdot P_{f} \right)}{\left(c_{c} \cdot \Delta T_{c} \right)} \tag{6.12}$$

The result passes to dependent calculations without being entered again.

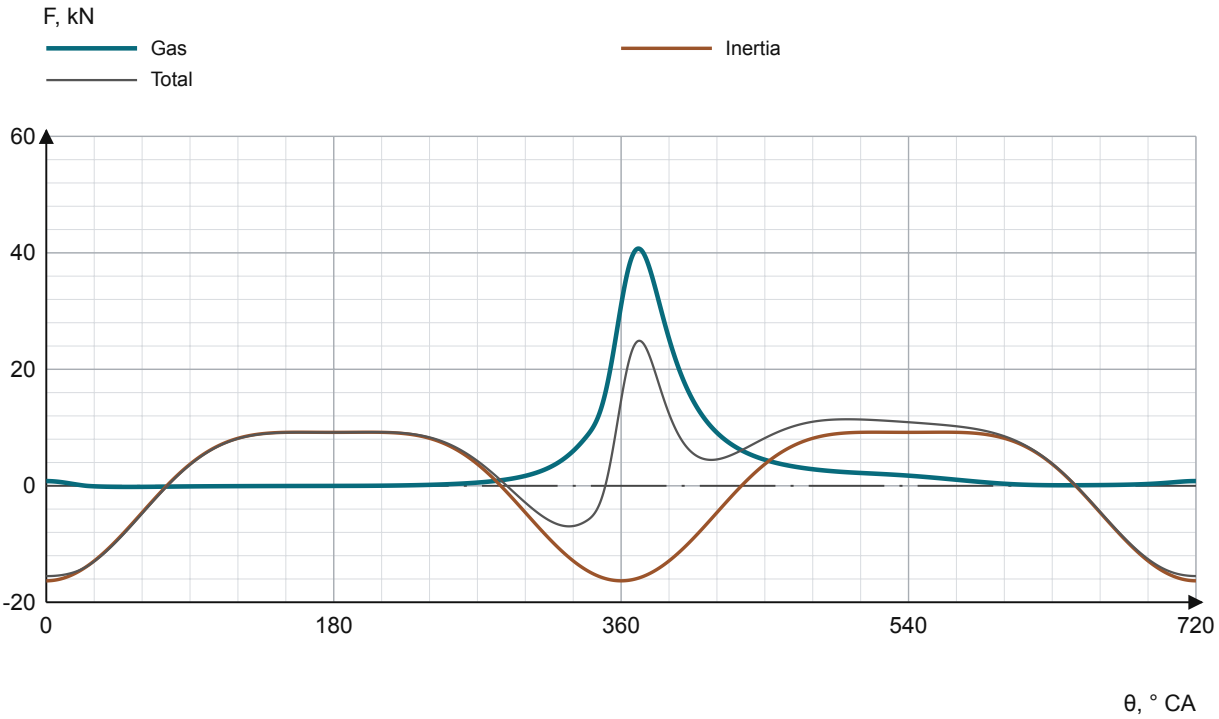


Illustration 6.1 — Axial forces of one cylinder. Petrol PFI · 110 kW / 7000 min⁻¹.

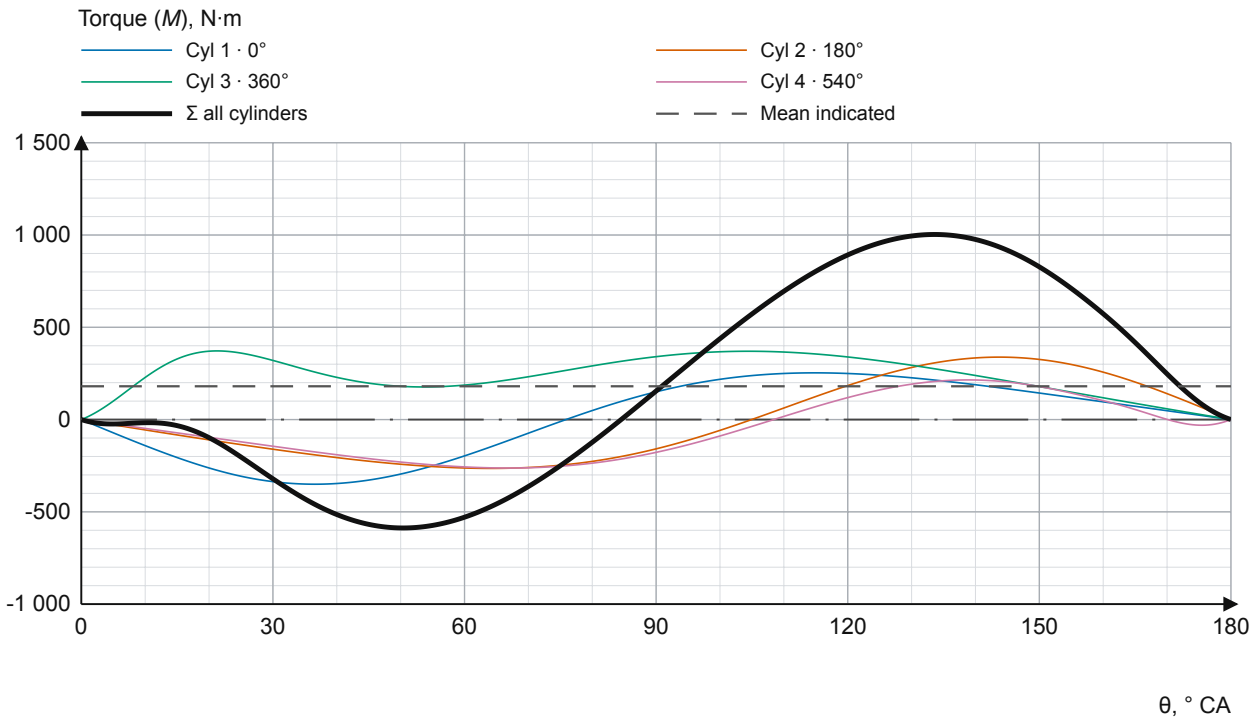


Illustration 6.2 — Cylinder contributions and total torque over one firing interval. Petrol PFI · 110 kW / 7000 min⁻¹.

7. Electric machine and inverter

Specified power (P) and rotational speed (n) determine required torque (M). Design mode produces a parametric characteristic; verification mode uses specified continuous power and torque. No electric-motor brand is required.

The operating envelope is limited by torque, power, phase current, and speed. Losses comprise copper heating in three phases and assumed rotational losses; the equation and symbol definitions are given below. The inverter is treated separately. This is an equivalent model, not a magnetic-circuit design.

A separate steady-state check of a permanent-magnet synchronous motor (PMSM) uses the direct d and quadrature q axes. Its parameters are inductances L_{d} and L_{q}, flux linkage ψ, and resistance R; current limits are peak amplitudes. It does not replace the equivalent model or design the windings or magnetic circuit.

S1 is continuous duty with adequate cooling; peak is short-duration duty with separate current and energy limits. Manufacturer ratings always depend on thermal conditions and the controller [emrax-ratings]. For N identical motors, the requirement is shared equally and the resulting torque and power are summed on an equivalent shaft.

6000 min⁻¹ in I-7 is an acceptable initial operating point, not a universal BEV speed limit. EMRAX 228 specifies peak power at 5500 min⁻¹; Brogen LCV-motor-2 lists 12000 min⁻¹ as peak speed. Base speed I-83 and maximum speed I-84 are adopted separately [emrax-ratings] [brogen-rpm].

Winding copper losses depend on phase resistance and RMS phase current. Rotational losses are assumed proportional to speed squared:

$$P_{\mathrm{loss}} = 3 \cdot R_{\mathrm{ph}} \cdot {I_{\mathrm{ph}}}^{2} + P_{\mathrm{rot,base}} \cdot \left(\frac{n}{n_{\mathrm{base}}} \right)^{2}$$

Here P_{loss} — losses of one motor, W; R_{ph} — single-phase resistance, Ω; I_{ph} — RMS phase current, A; $P_{\mathrm{rot,base}}$ — assumed rotational losses at base speed, W; n and n_{base} — current and base speeds in matching units. The factor 3 accounts for three phases. Inverter losses are included separately.

Sharing total torque between identical motors.

where P — Result of the corresponding preceding module; ω_{req} — Angular speed of the specified output shaft, rad/s; N_{em} — Number of identical motors

$$M_{\mathrm{m,req}} = \frac{P}{\left(\omega_{\mathrm{req}} \cdot N_{\mathrm{em}} \right)} \tag{7.1}$$

The result passes to dependent calculations without being entered again.

Phase current at the required point.

where $M_{\mathrm{m,req}}$ — Required torque per motor, N·m; k_{T} — Effective torque / phase-current coefficient, N·m/A

$$I_{\mathrm{ph}} = \frac{M_{\mathrm{m,req}}}{k_{\mathrm{T}}} \tag{7.2}$$

The result passes to dependent calculations without being entered again.

Resistive losses in three phases.

where R_{ph} — Single-phase resistance at the specified temperature, Ω; I_{ph} — Phase current at the required point, A

$$P_{\mathrm{Cu}} = 3 \cdot R_{\mathrm{ph}} \cdot {I_{\mathrm{ph}}}^{2} \tag{7.3}$$

The result passes to dependent calculations without being entered again.

Assumed rotational losses.

where $P_{\mathrm{fe,base}}$ — Rotational losses at base speed, W; n_{req} — Speed at which power I-6 is required, s⁻¹; n_{base} — Traction-motor base speed, s⁻¹

$$P_{\mathrm{rot}} = P_{\mathrm{fe,base}} \cdot \left(\frac{n_{\mathrm{req}}}{n_{\mathrm{base}}} \right)^{2} \tag{7.4}$$

The result passes to dependent calculations without being entered again.

Required DC power before the inverter.

where P — Result of the corresponding preceding module; P_{Cu} — Resistive losses in three phases, W; P_{rot} — Assumed rotational losses, W; η_{inv} — Inverter efficiency

$$P_{\mathrm{DC,req}} = \frac{(P + P_{\mathrm{Cu}} + P_{\mathrm{rot}})}{\eta_{\mathrm{inv}}} \tag{7.5}$$

The result passes to dependent calculations without being entered again.

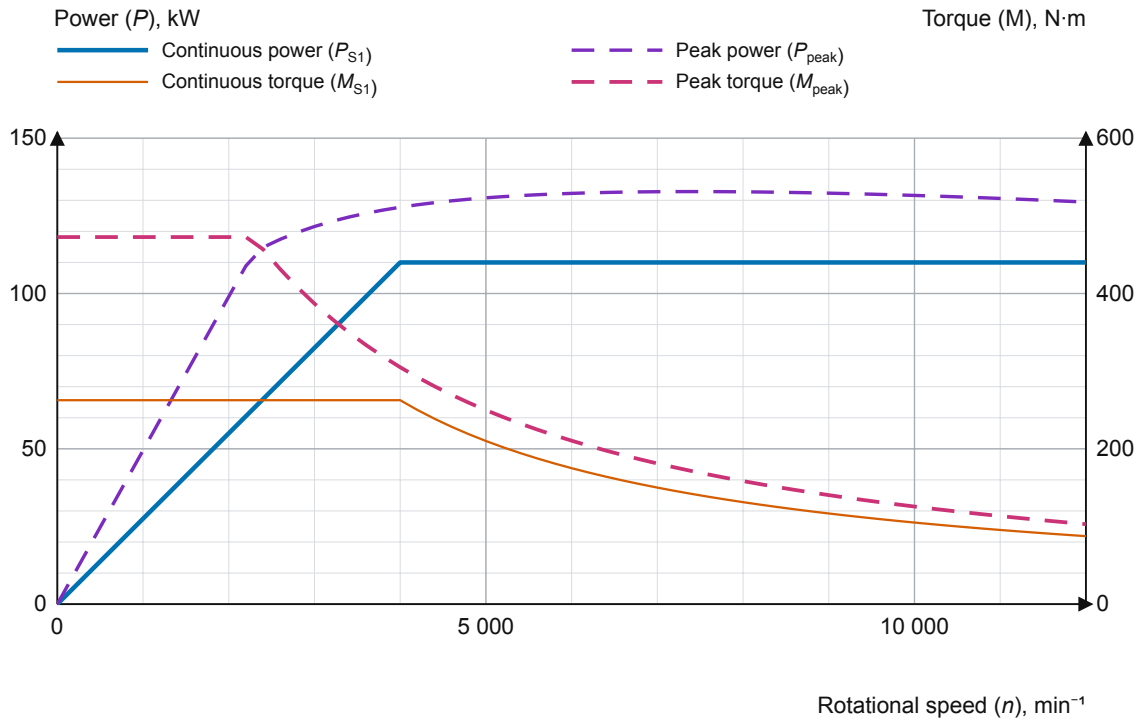
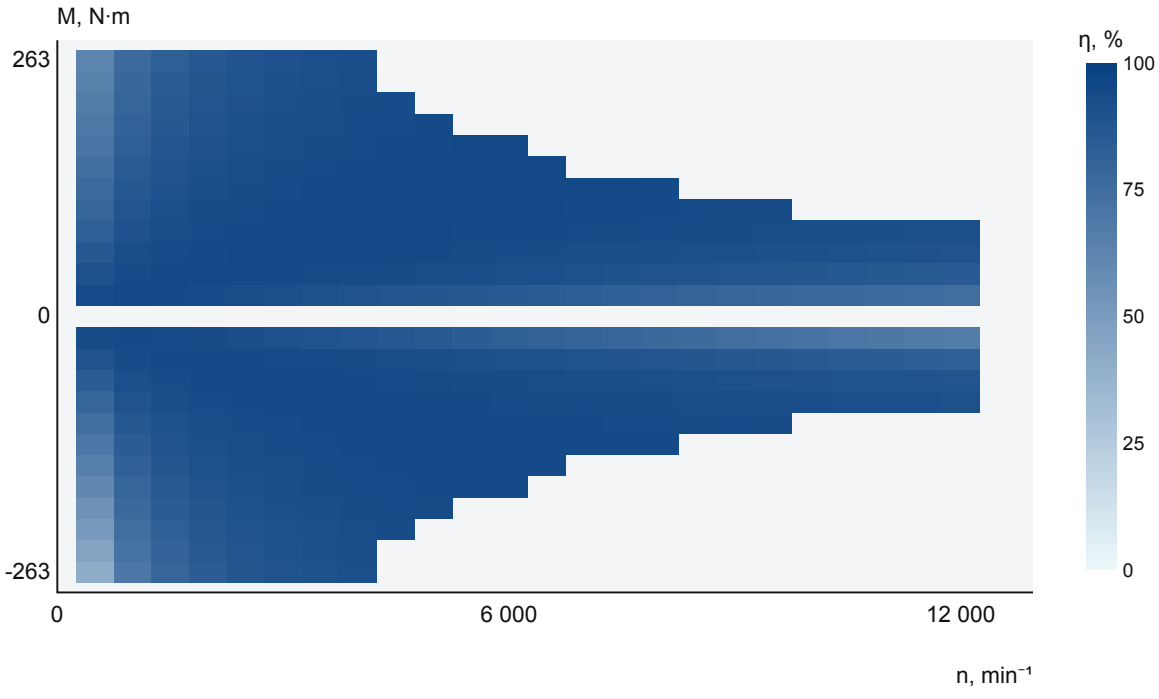


Illustration 7.1 — Electric motors: S1 and peak, torque and power. BEV · 110 kW · battery sizing for 250 km WLTC.



Light area: missing value or outside the operating envelope.

Electric-machine efficiency only; excludes the inverter. Modelled, not measured.

Illustration 7.2 — Modelled motor efficiency map, excluding the inverter. BEV · 110 kW · battery sizing for 250 km WLTC.

8. Battery: power, energy, and duration

Internal-resistance model (R_{int}, internal resistance) represents the battery as an open-circuit-voltage source (OCV, U_{oc}) and resistance R. Terminal voltage includes the resistive voltage drop; losses are I²R. The power equation selects the physical current root that approaches I = P/U_{oc} as R → 0. Current, voltage, and state-of-charge (SOC) limits apply simultaneously; power deficits are retained.

Design sequence: vehicle class and required occupants/payload → base mass and permissible gross mass → battery mass budget → energy, pack count, or target range → route → currents, consumption, and attainable range. Separate inputs I-105 and I-106 are retained for an educational constant-load check; its results do not replace the route calculation.

In the main calculation with a specified charge reserve, open-circuit voltage, resistance, and temperature are constant. A separate termination procedure based on loss of speed uses editable voltage and resistance dependencies on state of charge; its assumptions are explained in chapter 10. High-voltage equipment requires separate insulation, protection, cooling, and manufacturing checks. This document is not a battery assembly instruction.

The optional SOC/T model 0.7 is available in the separate Powertrain module: tabulated Voc, R, and current limits, Ah counting, and one thermal state. It requires an explicit data package; the basic Design module retains constant parameters rather than silently changing the method.

The additional electric-range calculation uses rolling resistance, aerodynamics, grade, and inertia from the adapted Automobile module. Energy capacity is specified directly, through the number of packs, or selected from the target range. The result depends on mass, speed, auxiliary load, and regeneration [automobile-ev-adaptation] [fastsim-method].

400 V is retained as the initial educational assumption for this power class. At the same power, doubling voltage approximately halves current, potentially reducing I²R losses and conductor requirements. 800 V requires all high-voltage components to be compatible and does not automatically double range. Comparing losses requires actual pack and winding parameters [porsche-800v].

I-96 provides coordinated 400 and 800 V presets and a custom value. BMW i4 xDrive40 specifies 398.5 V [bmw-i4-voltage]; Hyundai IONIQ 5 uses an 800 V architecture [hyundai-800v]. These illustrate different architectures. At the same power, doubling voltage approximately halves current, but actual losses also depend on resistance and layout. Changing voltage alone does not double energy or range.

The equation P = (Voc − IR)I gives the physical current root of the battery internal-resistance model. A negative discriminant means an unattainable operating point, not zero current.

where P — Result of the corresponding preceding module; U_{oc} — Equivalent battery open-circuit voltage, V; R_{int} — Equivalent battery resistance, Ω

$$I_{\mathrm{req}} = \frac{2 \cdot P}{\left(U_{\mathrm{oc}} + \sqrt{{U_{\mathrm{oc}}}^{2} - 4 \cdot R_{\mathrm{int}} \cdot P} \right)} \tag{8.1}$$

The result passes to dependent calculations without being entered again.

Energy is determined from the specified load duration and allowable SOC window. Energy capacity does not remove the battery current limit.

where U_{oc} — Equivalent battery open-circuit voltage, V; I_{req} — Battery current including internal resistance, A; t_{mission} — Duration of the separate constant-load check, s; SOC_0 — Initial state of charge; $\mathrm{SOC}_{\mathrm{min}}$ — Lower state-of-charge limit

$$E_{\mathrm{req}} = \frac{U_{\mathrm{oc}} \cdot I_{\mathrm{req}} \cdot t_{\mathrm{mission}}}{(\mathrm{SOC}_0 - \mathrm{SOC}_{\mathrm{min}})} \tag{8.2}$$

The result passes to dependent calculations without being entered again.

Equivalent charge capacity.

where E — Result of the corresponding preceding module; U_{oc} — Equivalent battery open-circuit voltage, V

$$Q_{\mathrm{nom}} = \frac{E}{U_{\mathrm{oc}}} \tag{8.3}$$

The result passes to dependent calculations without being entered again.

Electrical balance residual.

where C — Result of the corresponding preceding module; T — Result of the corresponding preceding module; H — Result of the corresponding preceding module

$$\Delta \mathrm{E} = C - T - H \tag{8.4}$$

The result passes to dependent calculations without being entered again.

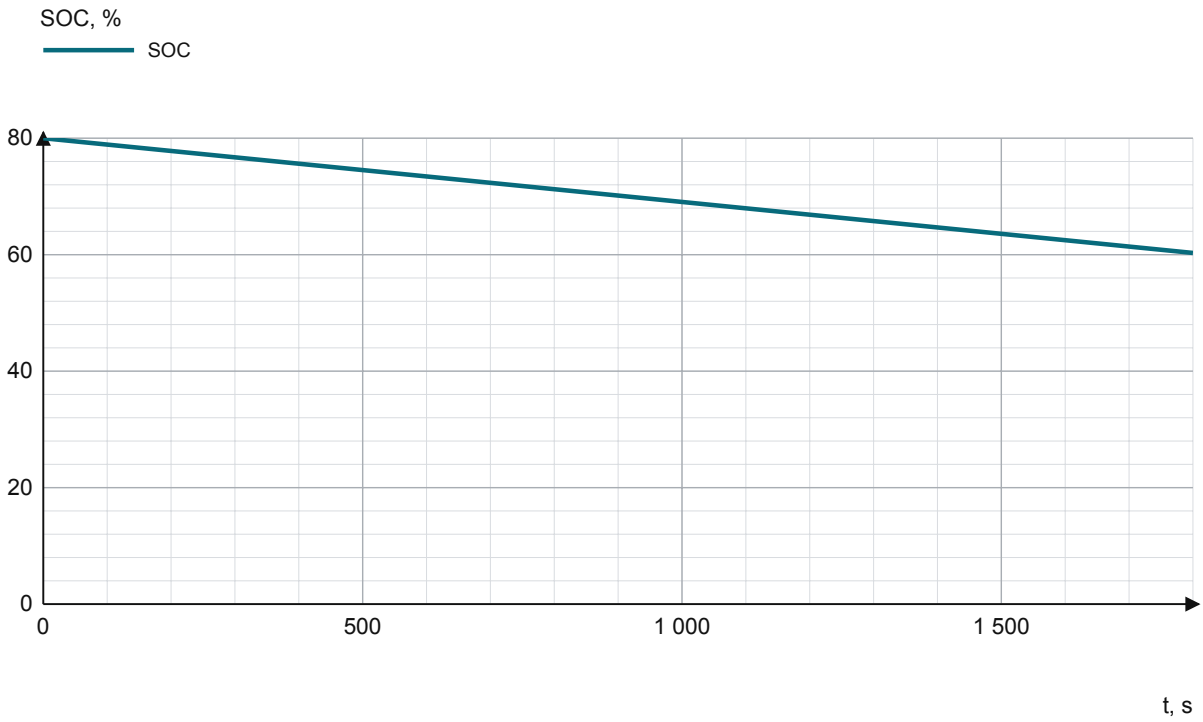


Illustration 8.1 — State of charge in the adopted energy scenario. BEV · 110 kW · battery sizing for 250 km WLTC.

9. Hybrid powertrain matching

Parallel P2: engine and motor torques are compared on the same shaft at the same speed. Series: the engine supplies energy through a generator to the DC bus; only the traction motor drives the output shaft.

The engine contributes calculated, rather than specified, power. Generator, electric-machine, inverter, and battery losses are included. Fuel and electricity are reported separately, together with SOC change; the model is not presented as a production controller.

Powertrain adds bounded terminal-SOC matching through the controller target SOC and sensitivity scenarios. Initial charge remains unchanged; failure or insufficient feasibility/data coverage does not justify comparing fuel consumption alone.

PHEV denotes the ability to charge from the grid, while series/parallel describes energy transfer. In P2, torques add at a common speed with the clutch engaged. In a series layout, engine power passes through the generator: generator and battery DC flows are summed, while the traction motor limits mechanical output. The PHEV electric range below is estimated with the engine switched off.

Consumption per 100 km is calculated for the selected route, gross mass, and explicit power split I-183/I-184. Current-engine fuel use and charge change are reported separately; this is one cycle without charge-sustaining correction or WLTP weighting.

Combined P2 power at the matched point and SOC_0.

where $P_{\mathrm{ICE,calc}}$ — Brake power after adopting the dimensions, W; M — Result of the corresponding preceding module

$$P_{\mathrm{calc}} = P_{\mathrm{ICE,calc}} + M \tag{9.1}$$

The result passes to dependent calculations without being entered again.

Generator-path power before limiting.

where $P_{\mathrm{ICE,calc}}$ — Brake power after adopting the dimensions, W; η_{gen} — Generator-path efficiency

$$P_{\mathrm{gen,raw}} = P_{\mathrm{ICE,calc}} \cdot \eta_{\mathrm{gen}} \tag{9.2}$$

The result passes to dependent calculations without being entered again.

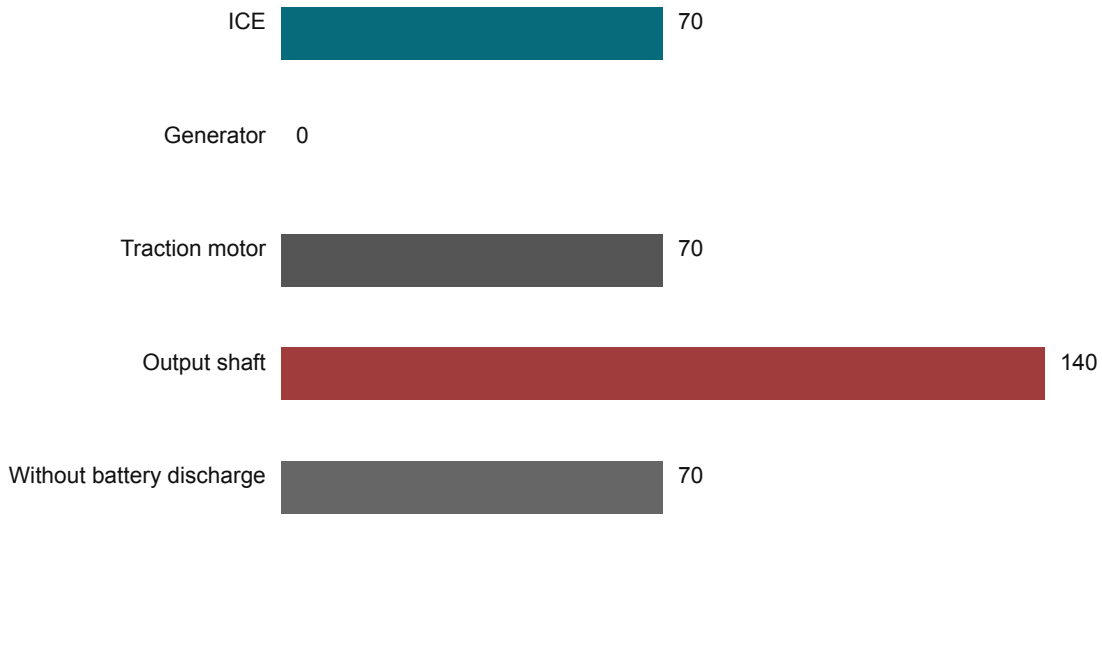


Illustration 9.1 — Flows at the calculation point, kW. PHEV P2 · 70 kW ICE + 70 kW electric traction.

10. Vehicle: speed, energy consumption, and range

BEV and hybrid electric-mode calculations use the current calculated electric machine, battery, and gross mass including payload. Characteristics from another vehicle are not substituted.

Maximum speed is limited by motor speed, reduction ratio, and available continuous power. Range is stated for the selected mode and SOC window; battery energy and estimated grid energy are shown separately.

Current output-shaft rotational speed.

where v — Check speed, m/s; i_{t} — Output-shaft-to-wheel ratio; r_{w} — Dynamic wheel radius, m

$$n_{\mathrm{shaft}} = \frac{v \cdot i_{\mathrm{t}}}{(\,2 \cdot \pi \cdot r_{\mathrm{w}}\,)} \tag{10.1}$$

The result passes to dependent calculations without being entered again.

Available tractive power at the wheels.

where P — Result of the corresponding preceding module; η_{t} — Driveline efficiency to the wheels

$$P_{\mathrm{w,av}} = P \cdot \eta_{\mathrm{t}} \tag{10.2}$$

The result passes to dependent calculations without being entered again.

Normal component of weight.

where m_{v} — Base vehicle mass according to I-168, kg; g — Result of the corresponding preceding module; i — Longitudinal grade

$$N = \frac{m_{\mathrm{v}} \cdot g}{\sqrt{1 + i^2}} \tag{10.3}$$

The result passes to dependent calculations without being entered again.

Rolling resistance.

where N — Normal component of weight, N; f_0 — Rolling-resistance coefficient

$$F_{\text{roll}} = N \cdot f_0 \tag{10.4}$$

The result passes to dependent calculations without being entered again.

Grade resistance force.

where N — Normal component of weight, N; i — Longitudinal grade

$$F_{\text{grade}} = N \cdot i \tag{10.5}$$

The result passes to dependent calculations without being entered again.

Aerodynamic drag.

where rho — Result of the corresponding preceding module; $C_{\text{d A}}$ — Drag coefficient × frontal area, m²; v — Check speed, m/s

$$F_{\text{air}} = \frac{\text{rho} \cdot C_{\text{d A}} \cdot v^2}{2} \tag{10.6}$$

The result passes to dependent calculations without being entered again.

Required wheel force.

where F_{roll} — Rolling resistance, N; F_{grade} — Grade resistance force, N; F_{air} — Aerodynamic drag, N

$$F_{\text{req}} = F_{\text{roll}} + F_{\text{grade}} + F_{\text{air}} \tag{10.7}$$

The result passes to dependent calculations without being entered again.

Required wheel power.

where F_{req} — Required wheel force, N; v — Check speed, m/s

$$P_{\text{w}} = F_{\text{req}} \cdot v \tag{10.8}$$

The result passes to dependent calculations without being entered again.

Supporting output-shaft demand with margin.

where P_{w} — Required wheel power, W; k — Result of the corresponding preceding module; η_{t} — Driveline efficiency to the wheels

$$P_{\mathrm{advised}} = \frac{P_{\mathrm{w}} \cdot k}{\eta_{\mathrm{t}}} \tag{10.9}$$

The result passes to dependent calculations without being entered again.

11. EPA maps and driving scenarios

The US Environmental Protection Agency (EPA) publishes powertrain test-data packages and maps. In this chapter, EPA denotes the source of reference maps, not certification of this calculation.

A fuel-consumption map describes $\dot{m}f(n,M)$, not an indicator diagram. It does not replace the geometry, temperatures, or forces of the current engine cycle. Component identity, fuel, operating envelope, units, and missing values are preserved [epa-intake].

Interpolation at the check point is restricted to the valid data domain. Empty and negative source cells do not become zero consumption. A map does not replace the design engine unless explicitly selected.

The current BEV vehicle and range are calculated in chapter 10. This chapter shows a separate external map for component comparison; it does not replace the calculated motor or complete vehicle.

The Bolt electric map represents motoring losses at 350 V. Inverter behaviour, a regeneration map, and verified continuous-current limits are not invented. It is used for the check point and plot; the general BEV continues to use the specified equivalent electric machine.

Mechanical power at the check point. [6]

where n_{map} — Map check-point speed, s⁻¹; M_{map} — Map check-point torque, N·m

$$P_{\mathrm{map}} = 2 \cdot \pi \cdot n_{\mathrm{map}} \cdot M_{\mathrm{map}} \tag{11.1}$$

A common check point is needed to convert the map consistently into efficiency and specific consumption.

Specific consumption from the EPA map. [6]

where f — Result of the corresponding preceding module; P_{map} — Mechanical power at the check point, W

$$b_{\mathrm{map}} = \frac{f}{P_{\mathrm{map}}} \tag{11.2}$$

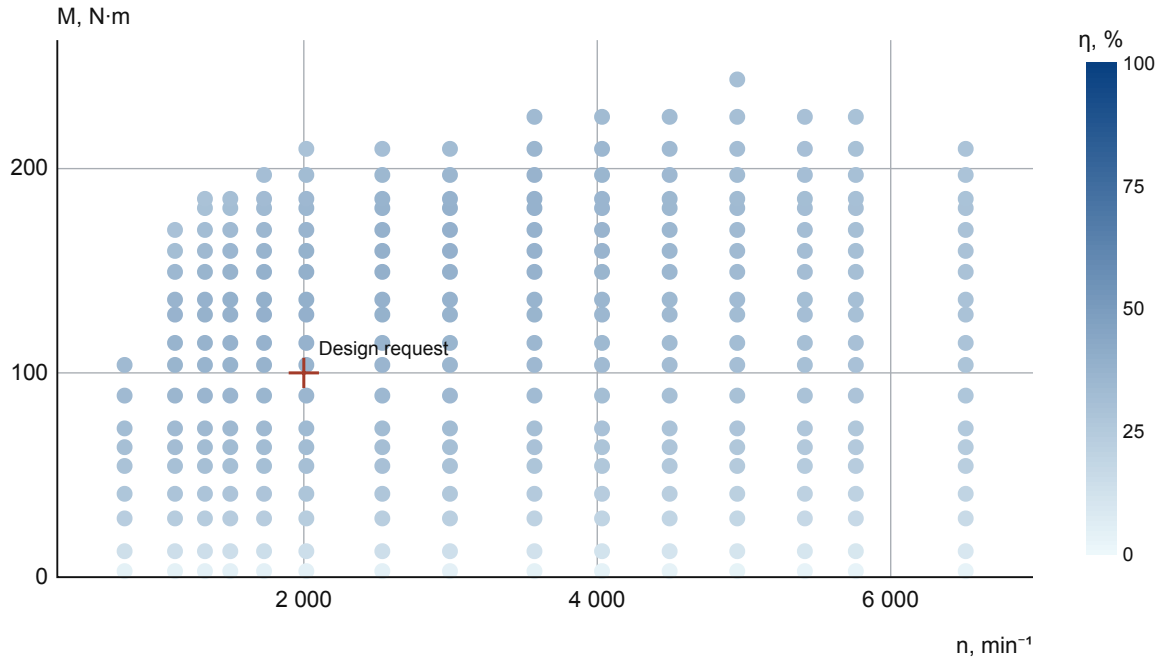
A missing map point has no numerical value and is not interpolated beyond the operating envelope.

Brake thermal efficiency at the EPA point. [6]

where P_{map} — Mechanical power at the check point, W; f — Result of the corresponding preceding module; H — Result of the corresponding preceding module

$$\eta_{\text{map}} = \frac{P_{\text{map}}}{(f \cdot H)} \tag{11.3}$$

The fuel heating value comes from the specific map package, not from another engine's fuel.



Known external-component points; the right-hand scale shows efficiency η, %.

No interpolation across missing data. The cross marks the design request.

Illustration 11.1 — External-component efficiency map: known points and design request. Petrol PFI · 110 kW / 7000 min⁻¹.

12. Strength and fatigue life: first approximation

Loads are transferred from the current cycle. The connecting rod is a bar; the piston pin is a simply supported beam; the shaft is a circular section in torsion. Default materials, concentration factors, and geometry are illustrative and editable [first-order-assumptions].

Goodman uses the stress amplitude and mean of the complete block; a beneficial compressive mean is not credited. Euler applies only when critical stress is below the specified elastic limit. These are preliminary checks, not a spatial crankshaft analysis [first-order-theory].

S–N: logarithmic interpolation between two entered reference points. One block repetition is 360° or 720° according to the selected cycle. Life is not extrapolated beyond the S–N interval; only a bound or insufficient data is stated. There is no automatically verified material fatigue curve, rainflow counting, or independent fatigue-life validation.

Equivalent amplitude under the adopted linear Goodman criterion, without credit for compressive mean stress.

> Kf is the assumed stress-concentration factor; σa and σm are stress amplitude and mean, Pa; σu is ultimate strength, Pa. The denominator must be positive.
>
> $\sigma_{eq} = K_f \sigma_a / (1 - K_f \max(\sigma_m,0)/\sigma_u)$ (12.9)

Life is determined by interpolation only between the two entered S–N reference points.

> N_1 and N_2 are specified cycle counts; σ_1 and σ_2 are their amplitudes, Pa; n is engine speed, min^{-1}; t is illustrative duration, h. One block is the selected 360° or 720° cycle, not a load spectrum. An exact N is not returned outside the reference interval.
>
> $b = \ln(\sigma_2/\sigma_1)/\ln(N_2/N_1)$; $N = N_1 (\sigma_{eq}/\sigma_1)^{(1/b)}$; $t = N/(30n)$ (12.10)

Nominal connecting-rod stress amplitude. [12]

> where hi — Result of the corresponding preceding module; lo — Result of the corresponding preceding module; A_{rod} — Connecting-rod section area, m²
>
> $$\sigma_{\mathrm{a}} = \frac{(\mathrm{hi} - \mathrm{lo})}{(2 \cdot A_{\mathrm{rod}})} \quad (12.1)$$
>
> This is an initial estimate under explicit assumptions, not manufacturing approval.

Mean axial stress, positive in tension. [12]

where hi — Result of the corresponding preceding module; lo — Result of the corresponding preceding module; A_{rod} — Connecting-rod section area, m²

$$\sigma_{\mathrm{m}} = \frac{-\left(\mathrm{hi} + \mathrm{lo}\right)}{\left(2 \cdot A_{\mathrm{rod}}\right)} \tag{12.2}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Goodman endurance utilisation. [12]

where K_{f} — Concentration / fatigue factor; σ_{a} — Nominal connecting-rod stress amplitude, Pa; S_{e} — Adopted component endurance limit, Pa; m — Result of the corresponding preceding module; S_{u} — Ultimate tensile strength, Pa

$$U_{\mathrm{G}} = \frac{K_{\mathrm{f}} \cdot \sigma_{\mathrm{a}}}{S_{\mathrm{e}}} + \frac{K_{\mathrm{f}} \cdot m}{S_{\mathrm{u}}} \tag{12.3}$$

U_{G} ≤ 1 passes only this simplified criterion. Mean compressive stress is taken as zero for this criterion.

Euler critical load of the weak section. [12]

where E — Young's modulus, Pa; I_{rod} — Weak-section second moment of area, m⁴; K_{L} — Effective-length factor; l — Connecting rod centre distance, m

$$F_{\mathrm{E}} = \frac{\pi^2 \cdot E \cdot I_{\mathrm{rod}}}{\left(K_{\mathrm{L}} \cdot l\right)^2} \tag{12.4}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Second moment of area of the piston-pin section. [12]

where d_{p} — Piston-pin outer diameter, m; d_{pi} — Piston-pin inner diameter, m

$$I_{\mathrm{p}} = \frac{\pi \cdot \left({d_{\mathrm{p}}}^4 - {d_{\mathrm{pi}}}^4\right)}{64} \tag{12.5}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Piston-pin bending stress: central load. [12]

where K_{f} — Concentration / fatigue factor; F — Result of the corresponding preceding module; l_{p} — Calculated piston-pin span, m; d_{p} — Piston-pin outer diameter, m; I_{p} — Second moment of area of the piston-pin section, m⁴

$$\sigma_{\mathrm{p}} = \frac{K_{\mathrm{f}} \cdot F \cdot l_{\mathrm{p}} \cdot d_{\mathrm{p}}}{\left(8 \cdot I_{\mathrm{p}} \right)} \tag{12.6}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Mean projected piston-pin pressure. [12]

where F — Result of the corresponding preceding module; d_{p} — Piston-pin outer diameter, m; b_{p} — Projected contact length, m

$$p_{\mathrm{p}} = \frac{F}{\left(d_{\mathrm{p}} \cdot b_{\mathrm{p}} \right)} \tag{12.7}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Torsion of an equivalent solid section. [12]

where K_{f} — Concentration / fatigue factor; M — Result of the corresponding preceding module; d_{s} — Equivalent shaft-section diameter, m

$$\tau_{\mathrm{s}} = \frac{16 \cdot K_{\mathrm{f}} \cdot M}{\left(\pi \cdot {d_{\mathrm{s}}}^{3} \right)} \tag{12.8}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

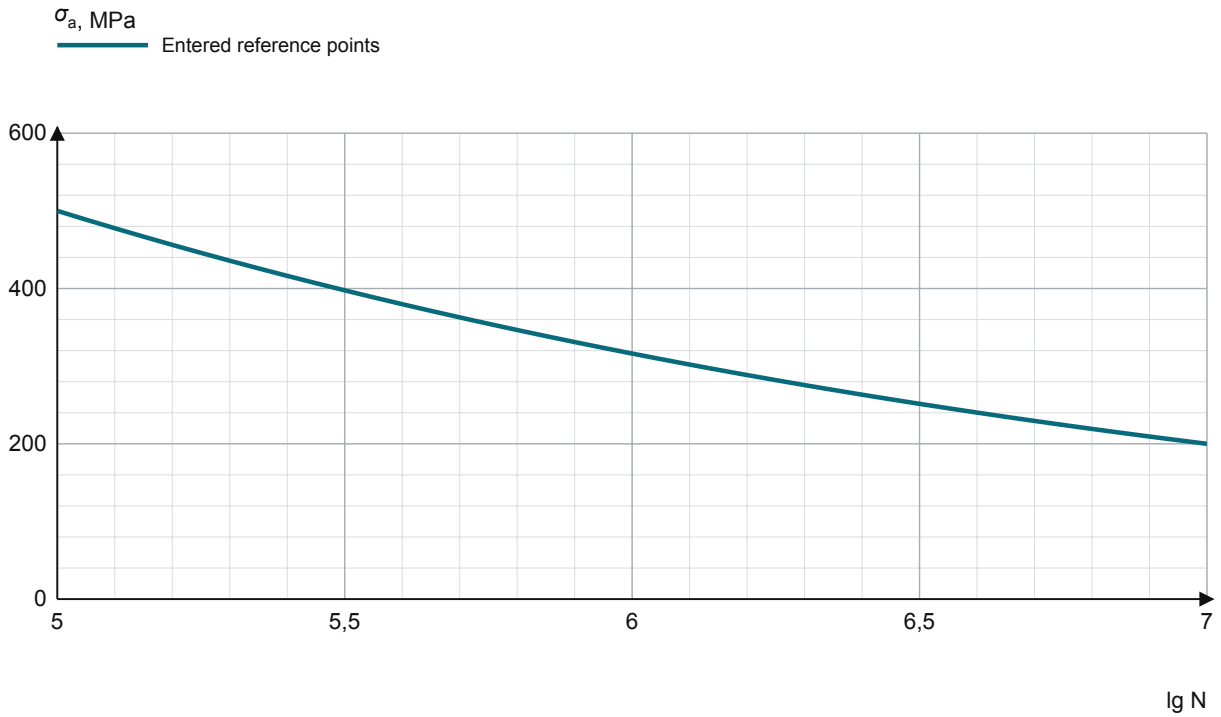


Illustration 12.1 — Illustrative S–N curve: only between entered reference points. Petrol PFI · 110 kW / 7000 min⁻¹.

13. Simplified control strategy

A first-order actuator is added to the traction source: τ dP/dt + P = P_{cmd}. An exact discrete solution for a constant command is used; τ and load fraction are editable educational assumptions. The transient plot is a separate check and does not modify current engine-cycle pressure.

The driving cycle uses the Powertrain limiting strategy: traction, battery charge/discharge, regeneration, P2, or series generation. A missing map point prevents reliable fuel-consumption calculation rather than disappearing from the balance. This algorithm is not ECU/BMS firmware [first-order-assumptions].

First-order actuator dynamics. The command is constant over the step; initially P=0.

> τ is the time constant, s; P_{cmd} — power command, W; t is time, s. The series uses the exact transition for a constant command.

τ dP/dt + P = Pcmd; P(t) = Pcmd (1 − exp(−t/τ)) (13.2)

Required power command. [12]

> where u — Requested load fraction; P_{calc} — Combined P2 power at the matched point and SOC_0, W

$$P_{\mathrm{cmd}} = u \cdot P_{\mathrm{calc}} \tag{13.1}$$

> This is an initial estimate under explicit assumptions, not manufacturing approval.

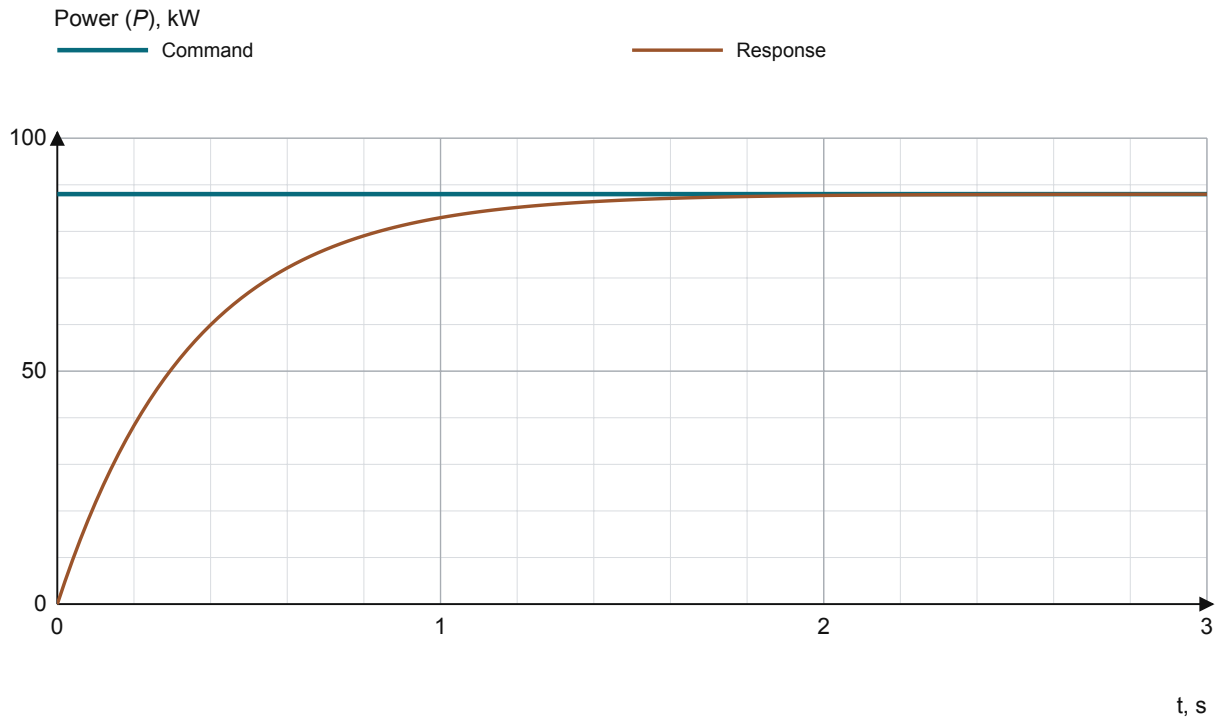


Illustration 13.1 — First-order power response. Petrol PFI · 110 kW / 7000 min⁻¹.

14. Electric machine: initial electromagnetic approximation

The steady-state permanent-magnet synchronous motor (PMSM) model uses direct d and quadrature q axes, inductances L_{d}, L_{q}, flux linkage ψ_{f} , and pole-pair count. Axis currents are peak amplitudes; the main project's phase-current inputs are root-mean-square (RMS) values. Space-vector pulse-width modulation (SVPWM) defines the ideal linear phase-voltage limit; no margin is modelled [first-order-theory].

A bounded id-grid search minimises current magnitude for the required torque. This is neither global optimisation nor finite-element analysis. Iron losses, saturation, demagnetisation, and winding temperature require further assessment.

The size estimate assumes τgap = Bδ Aline/2 and M = π D² L τgap/2. This is an explicit sinusoidal-approximation coefficient; windings, slots, and the magnetic circuit are not designed. Illustrative Bδ and Aline are not attributed to EMRAX or Bolt.

Bounded steady-state PMSM dq check with specified constant parameters.

> p is pole-pair count; ψ is flux linkage, Wb; Ld/Lq are inductances, H; Rs is phase resistance, Ω. id/iq and vd/vq use peak-amplitude Park normalisation. The operating point is searched on a finite id grid; failure to find one does not prove general impossibility.
>
> ωe=pωm; T=1.5p[ψ+(Ld−Lq)id]iq; vd=Rs id−ωeLq iq; vq=Rs iq+ωe(Ld id+ψ) (14.6)

Maximum phase-current amplitude. [12]

> where $I_{\mathrm{ph,lim}}$ — Maximum continuous phase current per motor, A
>
> $$I_{\mathrm{pk}} = \sqrt{2} \cdot I_{\mathrm{ph,lim}} \tag{14.1}$$
>
> This is an initial estimate under explicit assumptions, not manufacturing approval.

Maximum phase-voltage amplitude. [12]

> where U_{min} — Minimum terminal voltage, V
>
> $$U_{\mathrm{pk}} = \frac{U_{\mathrm{min}}}{\sqrt{3}} \tag{14.2}$$
>
> This is an initial estimate under explicit assumptions, not manufacturing approval.

Adopted mean air-gap shear stress. [12]

where B_δ — Adopted air-gap flux density, T; A_l — Linear current loading, A/m

$$\tau_\mathrm{gap} = \frac{B_\delta \cdot A_\mathrm{l}}{2} \tag{14.3}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Approximate active diameter. [12]

where $M_\mathrm{m,req}$ — Required torque per motor, N·m; k_L — Active-length-to-diameter ratio; τ_gap — Adopted mean air-gap shear stress, Pa

$$D_\mathrm{m} = \left(\frac{2 \cdot M_\mathrm{m,req}}{\left(\pi \cdot k_\mathrm{L} \cdot \tau_\mathrm{gap} \right)} \right)^{\left(\frac{1}{3} \right)} \tag{14.4}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

Approximate active length. [12]

where k_L — Active-length-to-diameter ratio; D_m — Approximate active diameter, m

$$L_\mathrm{m} = k_\mathrm{L} \cdot D_\mathrm{m} \tag{14.5}$$

This is an initial estimate under explicit assumptions, not manufacturing approval.

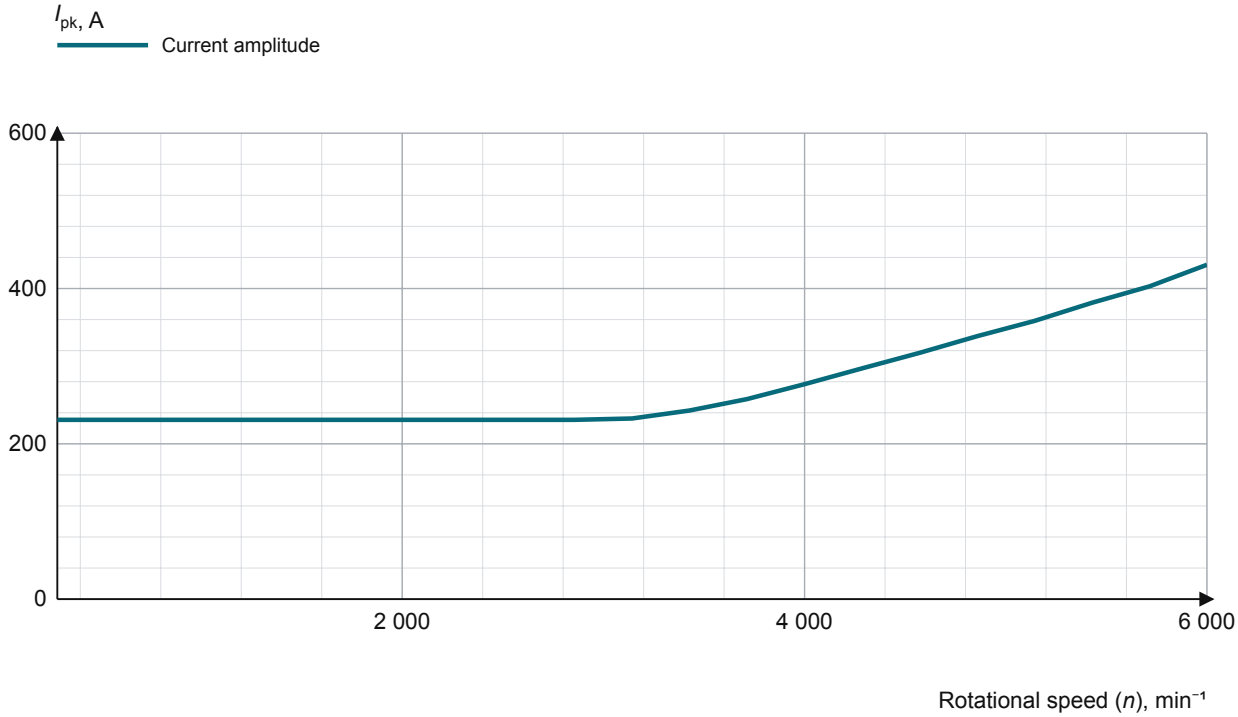


Illustration 14.1 — Required current in the dq check. BEV · 110 kW · battery sizing for 250 km WLTC.

15. Comparison with measurements

Paired CSV import accepts observed and predicted values with time, trip, and units. RMSE, bias, and maximum absolute error are calculated only from supplied pairs. Empty cells are excluded and counted separately.

Training and validation trips do not overlap. Until the user's measurements are supplied, the status is Not experimentally validated. The demonstration CSV checks arithmetic, not engine accuracy. The user's OBD2 data are not mixed with independent torque measurements.

Calculate errors from supplied, already aligned pairs without fitting again.

> y is the supplied observation; ŷ is the supplied prediction; N is the number of valid pairs. Missing values are not filled with zeros. Error units match the channel. Measurement provenance and sample independence require separate verification.
>
> $e_i=\hat{y}_i-y_i$; $\text{bias}=\Sigma e_i/N$; $\text{RMSE}=\sqrt{(\Sigma e_i^2/N)}$; $e_{max}=\max|e_i|$ (15.1)

16. Conclusions, sources, and limitations

Mathematical tests and balances check the implementation. They are not experimental validation, test-bench calibration, or evidence of emissions compliance.

The original assignment, adopted assumptions, numerical results, and external sources remain distinct. HTML, PDF, CSV, and the native document are generated from the same state.

Conclusions concern the applicable checks for the current variant. Installed EPA reference maps are used only within their data domains. Independent testing, complete spatial strength analysis, verified fatigue life, and emissions require separate evidence.

Relative overall requirement residual.

where P_{calc} — Combined P2 power at the matched point and SOC_0, W; P_{req} — Required output-shaft power, W

$$\delta \mathrm{P} = \frac{\left(P_{\mathrm{calc}} - P_{\mathrm{req}} \right)}{P_{\mathrm{req}}} \tag{16.1}$$

The result passes to dependent calculations without being entered again.

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